MTH603 - Numerical Analysis (Quiz 3)

Question # 8 of 10 (Start time: 03:06:00 PM, 24 June 2025)

Given the following data x:1 2 5 y:1 4 10 Value of 1st order divided difference f[2 , 5] is

Select the correct option

◉	2
○	1
○	0
○	-2

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Question # 6 of 10 (Start time: 03:05:20 PM, 24 June 2025)

Lagrange's interpolation formula is used when the values of the independent variable are

Select the correct option

◉	Not equally spaced
○	None
○	Constant
○	Equally spaced

MTH603 - Numerical Analysis (Quiz 3) Quiz Start

Question # 1 of 10 (Start time: 03:03:26 PM, 24 June 2025)

$For\ the\ given\ data$ points $(2,5)$, $(4,7)$, and $(6,9)$, $the\ zero-order\ divide\ difference\ will\ be$

Select the correct option

- [ ] 1
- [ ] 2
- [x] 5
- [ ] 0

MTH603 - Numerical Analysis (Quiz 3) Quiz

Question # 7 of 10 (Start time: 03:05:34 PM, 24 June 2025)

$For\ the\ given\ data$ points $(4,45)$, $(5,104)$, and $(6,190)$, the first $-$ $order\ divide\ difference\ will\ be$

Select the correct option

- [ ] 82
- [ ] 76
- [ ] none
- [x] 59

Click to Save

MTH603 - Numerical Analysis (Quiz 3)

Question # 2 of 10 (Start time: 03:03:38 PM, 24 June 2025)

Let

$$f(x, y) = 2x^3 + 6y^3 + 9xy$$

For x=0, 1,2,3,4, and y=0, 1,2,3,4 Then computing the value of f(2.5,3.5) is an example of .....

Select the correct option

○	Interpolation in one dimension
○	Interpolation in four dimensions
○	Interpolation in three dimensions
●	Interpolation in two dimensions

MTH603 - Numerical Analysis (Quiz 3) Qu

Question # 5 of 10 (Start time: 03:05:06 PM, 24 June 2025)

For the given four data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9	1.0
y	0.067	0.248	0.518	0.6812

Select the correct option

○	four
○	six
●	three
○	five

Click to S

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Question # 9 of 10 (Start time: 03:06:20 PM, 24 June 2025)

What will be the value of 'a' in the given divide difference table?

x	y	$1stD.D$	$2ndD.D$	$3rdD.D$
2	0.5	0.3	a	
4	1.1	0.3	-0.0125	-0.0021
6	1.7	0.25		
8	2..2			

Select the correct option

- ○ 0.0115
- ○ 0.0612
- ○ 0.0893
- ◉ 0

MTH603 - Numerical Analysis (Quiz 3) Quiz

Question # 3 of 10 (Start time: 03:03:54 PM, 24 June 2025)

If only two data points *are given, the formula for Lagrange's* interpolation *polynomial will be*

Select the correct option

- ◉ $y = f(x) = \frac{(x - x_1)}{(x_0 - x_1)} y_0 + \frac{(x - x_0)}{(x_1 - x_0)} y_1$
- ○ $y = f(x) = \frac{(x_1 - x_0)}{(x - x_0)} y_0 + \frac{(x_0 - x_1)}{(x - x_1)} y_1$
- ○ $y = f(x) = \frac{(x - x_0)}{(x_1 - x_0)} y_0 + \frac{(x - x_1)}{(x_0 - x_1)} y_1$
- ○ $y = f(x) = \frac{(x - x_0)}{(x_0 - x_1)} y_0 + \frac{(x - x_1)}{(x_1 - x_0)} y_1$

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Question # 4 of 10 (Start time: 03:04:20 PM, 24 June 2025) Total Marks: 1

For the given data points $(1, 0.3)$, $(3, 1)$, *and* $(5, 1.2)$ *the divide difference table will be given as*

Select the correct option

Reload Math Equations

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.125
6	1.2		

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.225
6	1.2		

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.525
6	1.2		

x	y	$1stD.D$	$2ndD.D$
2	0.3	0.35	
4	1	0.1	-0.0625
6	1.2		

Click to Save Answer & Move to Next Question

MTH603 - Numerical Analysis (Quiz 3) Quiz Start Time: 03:03 PM, 24 June 2025

Question # 10 of 10 (Start time: 03:06:59 PM, 24 June 2025) Total Marks: 1

For the giev three data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload Math Equations

$$y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.518) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.248)$$

$$y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$$

$$y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.518) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.067)$$

$$y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.248) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.067) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$$

Click to Save Answer & Move to Next Question